Reg. No: 1 9 2 5 2 4 1 4 4

SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name of the Student : S. MAGALAKSHMI

Programme: **B.E / B.Tech**

Course Code: **CSA1519**

Course Name: **Cloud Computing and Big Data Analytics**

Duration: **1:30 Hr**

Slot : D

Faculty : Dr.Arul Raj A.M

Date: 03/11/2026

Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.
CO2	Analyze virtualization architectures to identify performance and security issues in cloud computing environments.

Overall Marks Distribution :

Question No	Question	CO	Bloom's Level	Marks	
1	Q1 – Cloud Evolution and Service Models	CO1	BL3	10	
2	Q2 – Server Consolidation Energy Saving	CO1	BL3	10	
3	Q3 – VM Over-Commitme Security	CO2	BL4	10	
4	Q4 – Data Centre, SDDC, and Privacy	CO2	BL4	10	
5	Q5 – Edge Filtering and Serverless Computing	CO2	CO2	10	
Total				50	

Student Signature	Faculty Signature	Program Director

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices. **(10) (CO1) (BL3)**

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25%
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65%
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service sharing between the universities. Examine the federation challenges and the privacy risks

associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price of ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

(10) (CO2) (BL3)

Given data

Parameter	**Value**
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25% of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

CIA-1

NAME : MAGALAKSHMI

REG No. 192524144

SUBJECT: Cloud Computing

DATE: 3/11/2026

1) Developments in Company Hardware

i) Virtualization

It allow multiple VM's to run on one physical Server

Advantages:

Better resource utilization

Multiple OS can run on Same Server

Easy management.

ii) Multi-Core processor:

Modern processors contain multiple CPU cores

Advantages:

Faster Processing

Improves performance.

iii) High-Capacity Storage:

Modern Storage device Contain more Storage.

Advantages:

Stores large amount of data.

Supports backup

Provides faster data access.

iv) High Speed Networking:

It allows users to access cloud applications quickly from different locations.

Developments in Internet Software.

Web browsers allow users to access Cloud environments

Web Services support application Integration.

Cloud Service Models:

A) IaaS - Infrastructure as a Service

* IaaS provides virtual machines, Storage and networkings.
* The Company manages:
  - Operating System
  - Installed Software
  - Application
  - Configuration

The cloud provider manages physical infrastructure

B) PaaS - Platform as a Service

* PaaS provides a development and deployment platform.
* The Company manages application Code and data, while the Provider manages Servers, operating System and runtime environments.

C) SaaS - Software as a Service

- SaaS provides complete software over the internet.
- Users access the internet without installing the software or maintaining the underlying infrastructure.

D) XaaS Everything as a Service

XaaS is broad model in which different IT resources are provided as Services

Conclusion.

IaaS is suitable for migrating existing applications, PaaS is suitable for developing new applications and SaaS is suitable for using ready-made software and XaaS provides additional services based on organizational requirements.

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2) Virtualization Consolidation - Numerical Problem.

Given data:

Existing physical Servers = 15

CPU utilization per Server = 25%

Power Consumption per old Server = 350W

" " new host = 450W

Target CPU utilization per host = 65%

Fault tolerance = N+1

Electricity Cost = ₹ 9/KWh

Operating hours = 24 × 365

Assumption.

All 15 physical servers have equal CPU capacity, and their workloads are consolidated without loss of workload capacity.

a) Number of Active Hosts & Total Hosts.

Total Workload:

$$15 \times 25\% = 375\%$$

No. of active hosts:

$$N = \frac{375}{65} = 5.769$$

Rounding up:

$N = 6$ active hosts

Including $N+1$ fault tolerance:

$6 + 1 = 7$ total host.

b) Consolidation Ratio.

$$\text{Consolidation Ratio} = \frac{\text{Old Servers}}{\text{New Hosts}}$$

$$= \frac{15}{7}$$

$$2.14 : 1$$

If calculated using only six active hosts:

$$\frac{15}{6} = 2.5 : 1$$

c) CPU Utilization.

$$\frac{375}{7} = 53.57\%$$

When all seven hosts are active: 53.57%.

When one host fails, 8 x host remain:

$$\frac{375}{6} = 62.5\%$$

Both Values are within the 65% target

d) Annual Electricity Cost Saving.

Old System Power

$$15 \times 350 = 5250W = 5.25\ kW$$

New System power:

$$7 \times 450 = 3150W = 3.15\ kW$$

Annual Operating hours:

$$24 \times 365 = 8760$$

Old Annual electricity Cost:

$$5.25 \times 8760 \times 9$$

$$\Rightarrow ₹\ 413,910$$

New annual electricity cost:

$$3.15 \times 8760 \times 9$$

$$\Rightarrow ₹\ 248,346$$

Annual Saving:

$$413910 - 248346$$

$$₹\ 165,564$$

Conclusion:

Server Consolidation reduces the no. of physical Servers, Power Consumption & Operating Cost while maintaining N+1 fault tolerance.

5. Virtualization Host - CPU & Memory Sufficiency.

Given Data.

Physical CPU Cores = 24

Total RAM = 96 GiB

Hypervisor reservation = 8 GiB

Available RAM = 88 GiB

No. of VMs = 12

vCPU per VM = 3

RAM per VM = 8 GiB

a) Over-Commitment Ratios.

$12 \times 3 = 36$ vCPUs

$\frac{36}{24} = 1.5 : 1$

Total allocated RAM:

$12 \times 8 = 96$ GiB

Memory Over-Commitment ratio:

$\frac{96}{88} = 1.09 : 1$

b) Memory Sufficiency

Each VM uses 5 GiB

$12 \times 5 = 60$ GiB

Available memory = 88 GiB

$88 - 60 = 28$ GiB

Peak Operation

Eight VMs use 8 GiB each:

$8 \times 8 = 64$ GiB

Four VMs

$4 \times 6 = 24$ GiB

Total peak memory:

64 + 24 = 88 GB

c) CPU Sufficiency

Total allocated vCPU = 36

Each VM uses 35% of its allocated CPU

36 × 0.35 = 12.6

Physical CPU cores = 24

$\frac{12.6}{24} \times 100 = 52.5\%$

Peak Operation:

All VMs use 75%

36 × 0.75 = 27

$\frac{27}{24} \times 100 = 112.5\%$

Excess demand:

27 - 24 = 3 Cores

CPU is insufficient during peak operations.

d) Performance & Security Issues.

CPU Contention → Add CPU cores or migrate VMs

Memory pressure → Add RAM or use memory optimization

Unauthorized access → Implement MFA & RBAC

4. Cloud Federation Between Two Universities.

A. Conventional Cloud Data Center Architecture

Each University maintains:

Physical Servers

Storage Systems.

Network infrastructure

Hypervisors.

The universities share selected services through secure communication channels.

B. Software - Defined Data Center.

SDDC provides software - controlled infrastructure.

It supports.

- Dynamic allocation of computing resources.
- Automated VM Provisioning
- Flexible storage allocation.

c) Role of SOA

SOA enables independent services to communicate

d) Federation Challenges.

Different cloud platforms → Standardized APIs

Privacy risks → Data minimization.

Resource Sharing → Federation agreements.

e) Privacy Risks.

Federated identity : User identity information is shared between Universities. Incorrect access control may expose personal information.

Safeguards:

Use SAML or Open ID Connect.

Apply least - privilege access.

Obtain User Consent where required.

Establish data - sharing agreements.

Edge filtering and Cloud Object Storage.

Given data.

No. of Cameras → 150

Video Size → 4MB

Interval → 3 min

Storage Cost → ₹ 1.50/GB·month

Serverless invocation cost → ₹0.00015

Filtering percentage → 25% Upload.

Month → 30 days.

a) Daily Data Volume without Filtering.

clips per camera per day:

$$\frac{24 \times 60}{3} = 480$$

Total clips per day:

$$150 \times 480 = 72000$$

Daily data Volume:

$$72000 \times 4 = 288000 \text{ MB}$$

288GB/day

Average bandwidth:

$$\frac{288000 \times 8}{86400}$$

26.67 Mbps

b) Storage cost without Filtering.

Monthly data:

$$288 \times 30 = 8640 \text{ GB}$$

Storage cost:

$$8640 \times 1.50$$

₹ 12,960

c) Edge Filtering

Uploaded data:

288×0.25

72 GB/day

Average bandwidth

26.67×0.25

6.67 Mbps

Monthly Storage

$72 \times 30 = 2160$ GB

Storage Cost:

2160×1.50

₹ 3,240

Monthly Storage Saving

$12960 - 3240$

₹ 9,720

d) Monthly Serverless Invocation Cost.

Daily uploaded clips:

$72000 \times 0.25 = 18000$

Monthly Invocations:

$18000 \times 30 = 540000$

Monthly invocation Cost:

540000×0.0015

₹ 81